the dresses that resembles the most a purchase, then we can't even guarantee the precision of our metric: maybe a couple of feature vectors is slightly closer than another pair, but in the end, to the human eye, the latter images look more similar!

So, as long as the approximation error ϵ is not too large, chances are that an approximated result for "most similar image" will be as good as, or maybe better, than the exact result.

The interested reader can find plenty of literature on the topic of approximated similarity search, and delve deeper into the concepts that we could superficially threat here. As a starting point I suggest, for instance, with the remarkable work of G. Amato[16].

10.5 Ss$^+$-Tree[17]

So far, we have used the original Ss-tree structure, as described in the original paper by White and Jain[18]; Ss-trees have been developed to reduce the node overlapping, and in turn the number of leaves traversed by a search on the tree, with respect to R-trees (and k-d trees).

10.5.1 Are Ss-trees Better?

With respect to k-d trees, the main advantage of this new data structure is that it is self-balancing, so much so that all leaves are at the same height. Also, using bounding envelopes instead of splits parallel to a single axis, mitigates